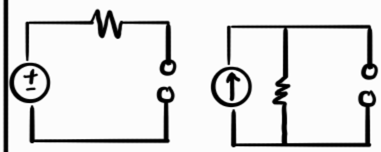
$$V_{out} = V_{in} \cdot \left(\frac{R_{out}}{R_{total}}\right)$$

$$I_{out} = I_{total} \cdot \left(\frac{R_{total}}{R_{out}}\right)$$

Thevenin | **Norton**



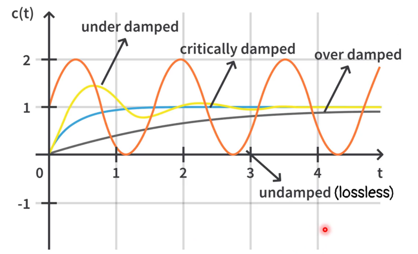
$$V_{oc} = I_{sc} R_{eq}$$

1^{st} order circuits [RC / LC circuit]

$\tau = RC$ or L/R [Steady state $t = 5\tau$]

$$V_c(t) = k_1 e^{-t/\tau} + \underbrace{k_2}_{V_c(\infty)}$$

$$i_L(t) = k_1 e^{-t/\tau} + \underbrace{k_2}_{i_L(\infty)}$$



2^{nd} order circuits (RLC circuits)

1) Make eqn only have 1 dependent & independent variable
2) Make it into 1^{st} order ODE w highest order coeff. 1.
3) Convert to Characteristic Eqn

$$V_L = L\frac{d\,i_L(t)}{dt}$$

$$I(t) = \frac{1}{L}\int_0^t V_L(t)\,dt + V_L(0)$$

$$i_c(t) = C\frac{dV_c(t)}{dt}$$

$$V_c(t) = \frac{1}{C}\int_0^t i_c(t)\,dt + i_c(0)$$

① $\frac{d^2 f(t)}{dt^2} + 2\alpha \frac{df(t)}{dt} + \omega_0^2 f(t) = 0$

Damping constant
1) Parallel RLC $= \frac{1}{2RC}$
2) Series RLC $= \frac{R}{2L}$
[Decay tendency]

Undamped Natural Freq.
$1/\sqrt{LC}$ for parallel & series RLC
[Oscillation tendency]

② $s^2 + 2\alpha s + \omega_0^2 = 0$; $s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 + \omega_0^2}$
[Characteristic Eqn]

③ $\frac{d^2 f(t)}{dt^2} + 2\omega_0 \zeta \frac{df(t)}{dt} + \omega_0^2 f(t) = 0$

Damping ratio
$\zeta = \alpha/\omega_0 = \frac{\sqrt{LC}}{2R}$
[Normalized damping]

④ $\frac{d^2 f(t)}{dt} + \frac{\omega_0}{Q}\frac{df(t)}{dt} + \omega_0^2 f(t) = 0$

Quality factor
Parallel $= \omega_0 RC$
Series $= \omega L/R$
[Energy-preserving tendency]

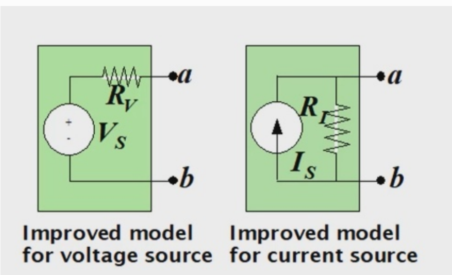
Parallel $\alpha = \frac{1}{2RC}$ Series $\alpha = R/2L$	OVERDAMPED	CRITICALLY DAMPED	UNDERDAMPED	LOSSLESS
Roots	Both real & negative	Equal, real & negative	Complex conjugate	Imaginary
s_1, s_2	$-\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	$-\alpha,\ -\alpha$	$-\alpha \pm j\omega_d$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	$\pm j\omega_0$
Damping const α	$\alpha > \omega_0$	$\alpha = \omega_0$	$0 < \alpha < \omega_0$	$\alpha = 0$
Quality factor $\omega_0/2\alpha$ Q	$0 < Q < 0.5$	$Q = 0.5$	$Q > 0.5$	$Q \to \infty$
Damping ratio α/ω_0 ζ	$\zeta > 1.0$	$\zeta = 1.0$	$0 < \zeta < 1.0$	$\zeta = 0$
homogenous response (series & parallel)	$k_1 e^{s_1 t} + k_2 e^{s_2 t}$	$(k_1 + k_2 t)e^{-\alpha t}$	$k_1 e^{-\alpha t}\cos(\omega_d t) + k_2 e^{-\alpha t}\sin(\omega_d t)$	

$$A\angle\phi = A\cos(\omega t + \phi) = Ae^{j\phi}$$

$$\sin\theta = \cos(\theta - 90°)$$

$$\cos\theta = \sin(\theta + 90°)$$

Source transformation



$Z = R + jX$ [Final ans : Complex form]

Capacitor, $Z_c = -j\frac{1}{\omega C}$

Inductor, $Z_L = j\omega L$

Advance AC S.S Analysis

① Mesh (loop w/ no other loop inside it) analysis

1) Draw mesh current [usually clockwise]
2) Write KVL for every mesh current
3) Solve simultaneous eqn

② Nodal Branch (≥3 wires connected to node) analysis

1) Find all node branch w/ more than 3 wires
2) Write KCL eqn for every node
3) Solve simultaneous eqn

③ Unit current method

1) Assume 1A flow through the wanted output
2) Then use $\frac{V_{output}}{V_{source}} = \frac{V_{output(unit)}}{V_{source(unit)}}$ @ $\frac{I_{output}}{V_{source}} = \frac{I_{output(unit)}}{V_{source(unit)}}$

$\frac{I_{output}}{I_{source}} = \frac{I_{output(unit)}}{I_{source(unit)}}$ @ $\frac{V_{output}}{I_{source}} = \frac{V_{output(unit)}}{I_{source(unit)}}$

Power Factor Correction

Without capacitor:

$$S_{old} = P_{old} + jQ_{old} = |S_{old}| \angle \theta_{old}$$
$$pf_{old} = \cos(\theta_{old})$$

With capacitor:

$$S_{new} = S_{old} + S_{capacitor} = P_{old} + jQ_{old} - jQ_{capacitor} = |S_{new}| \angle \theta_{new}$$
$$pf_{new} = \cos(\theta_{new})$$

$$Q_{capacitor} = |V_L||I_{capacitor}| = |V_L|^2 \omega C$$

$$\tan\theta_{new} = \frac{Q_{old} - Q_{capacitor}}{P_{old}}$$

+ve = consumed, -ve = supplied

$V_{rms} = \frac{V_m}{\sqrt{2}}$, $I_{rms} = \frac{I_m}{\sqrt{2}}$, $X_{rms} = \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} [x(t)]^2 dt}$

Instantaneous Power, $p(t) = \frac{V_m I_m}{2}\left[\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i)\right]$

Avg / Real Power, $P(t) = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) = \overbrace{(V_{rms})(I_{rms})}^{P_{apparent}} \cos\overbrace{(\theta_v - \theta_i)}^{pf,\ \theta_Z} = |S| \cos\theta_Z$

Max avg Power, $P_{max} = \frac{1}{2}\left[\frac{|V_{oc}|^2}{4 R_{thevenin}}\right]$, where $Z_{thevenin} = R_{thevenin} + jX_{thevenin}$

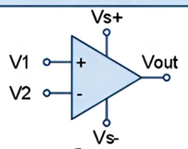
Complex Power, $S = (V_{rms})(I_{rms})^* = (V_{rms})(I_{rms}) \angle \overbrace{\theta_v - \theta_i}^{pf,\ \theta_Z} = \overbrace{P}^{\text{Real Power}} + j\overbrace{Q}^{\text{Reactive Power}}$

$|S| = \sqrt{P^2 + Q^2} = (V_{rms})(I_{rms}) = |I_{rms}|^2 Z$

$P = |I_{rms}|^2 R$, $Q = |I_{rms}|^2 X$

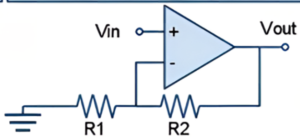
Basic Operational Amplifier Configurations

Voltage Comparator



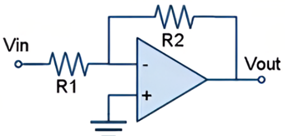
$V_{out} = \begin{cases} V_{s+} & V1 > V2 \\ V_{s-} & V1 < V2 \end{cases}$

Non-Inverting Amplifier



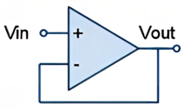
$A_v = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1}$

Inverting Amplifier



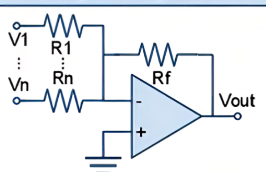
$A_v = \frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$

Voltage Follower



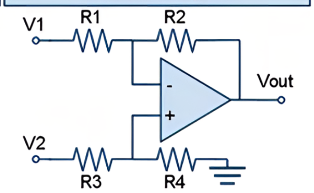
Vout = Vin

Inverting Summing Amplifier



$V_{out} = -R_f\left[\frac{V_1}{R_1} + \dots + \frac{V_n}{R_n}\right]$

Differential Amplifier

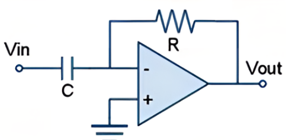


$V_{out} = \left[1 + \frac{R_2}{R_1}\right]\left[\frac{R_4}{R_3 + R_4}\right][V_2] - \left[\frac{R_2}{R_1}\right][V_1]$

If $R_1 = R_3$, $R_2 = R_4$

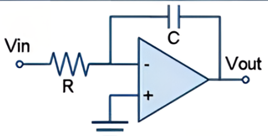
$V_{out} = \left[\frac{R_2}{R_1}\right][V_2 - V_1]$

Differentiator Amplifier



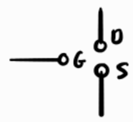
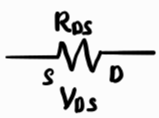
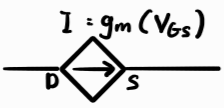
$V_{out} = -RC\left[\frac{dV_{in}}{dt}\right]$

Integrator Amplifier



$V_{out} = -\frac{1}{RC}\int V_{in}\, dt + V_{in}(0)$

MOSFET

$V_{GS} < V_{th}$ [Cutoff]	
$V_{GS} > V_{th}$ $V_{DS} < V_{GS} - V_{th}$ [Triode]	
$V_{GS} > V_{th}$ $V_{DS} > V_{GS} - V_{th}$ [Saturation]	

BJT

$V_{BE} < V_{th}$	$I_c = 0$	Cutoff
$V_{BE} \approx V_{th}$ $V_{CE} \geq V_{BE}$	$I_c = \beta I_B$	Forward Active
$V_{BE} \approx V_{th}$ $V_{CE} \leq V_{BE}$	$I_c < \beta I_B$	Saturation